数理方程

第13周
20260522

1. Spherical Bessel function
2. 特征线；2阶PDE分类与标准形式

极坐标：
- 角向：$\cos m\phi$, $\sin m\phi$
- 径向
  - Laplace eq.：幂函数 $r^{\pm m}$
  - Helmholtz eq.：Bessel $\frac{1}{r}\frac{d}{dr}\left(r\frac{dR}{dr}\right)+\left(k^2-\frac{m^2}{r^2}\right)R=0$

$\nabla^2 u+k^2u=0$ $\quad$ $\nabla^2 u+\lambda u=0$

球坐标：
- 角向：Legendre
- 径向
  - Laplace方程 $\Rightarrow$ 幂函数 r^l, $r^{-(l+1)}$
  - Helmholtz方程 $\Rightarrow$ 球Bessel函数

$\nabla^2 u+k^2u=0$

球Bessel

球坐标 Helmholtz equation 角向

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\left(k^2-\frac{l(l+1)}{r^2}\right)R=0$$

$l=0,1,2\cdots$

$$\Rightarrow \frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+[k^2r^2-l(l+1)]R=0$$

$x=kr$

$$\Longrightarrow \frac{d}{dx}\left(x^2\frac{dR}{dx}\right)+[x^2-l(l+1)]R=0 \leftarrow \text{球Bessel equation}$$

$x^2R''+2xR'+\cdots$

Bessel: $x^2R''+xR'+(x^2-\nu^2)R=0$

球Bessel eq. 和 Bessel eq. 比 xR' 项多因子 2

$$R(x)=a_0x^\rho+a_1x^{\rho+1}+\cdots \Rightarrow [\rho(\rho+1)-l(l+1)]x^\rho+\cdots=0$$

指标方程 $\rho(\rho+1)-l(l+1)=0 \Rightarrow \rho_1=l,\ \rho_2=-(l+1)$

$x^l,\ x^{-(l+1)}$

↓ 乘以 $\sqrt{x}$

$x^{l+\frac{1}{2}},\ x^{-(l+\frac{1}{2})}$

$x\to 0$ 时：$J_\nu\sim x^\nu$，$J_{-\nu}\sim x^{-\nu}$

$\Rightarrow$ 记 $\sqrt{x}R(x)=\sqrt{\frac{\pi}{2}}v(x)$，

即 $R(x)=\sqrt{\frac{\pi}{2x}}v(x)$，

希望与Bessel建立关系。

$\Rightarrow R' = \sqrt{\frac{\pi}{2}}\left(-\frac{1}{2}x^{-\frac{3}{2}}v + x^{-\frac{1}{2}}v\right)$

$(x^2R')' = \sqrt{\frac{\pi}{2}}\left(-\frac{1}{4}x^{-\frac{1}{2}}v + x^{\frac{1}{2}}v' + x^{\frac{3}{2}}v''\right) \Rightarrow x^{\frac{3}{2}}v'' + x^{\frac{1}{2}}v' - \frac{1}{4}x^{-\frac{1}{2}}v$

$+[x^2 - l(l+1)]x^{-\frac{1}{2}}v = 0$

$$x\frac{d}{dx}\left(x\frac{dv}{dx}\right) + \left[x^2 - \left(l+\tfrac{1}{2}\right)^2\right]v = 0$$

$l+\frac{1}{2}$ 阶 Bessel

$\Downarrow$

$v(x)$ 解 $J_{l+\frac{1}{2}}(x)$, $N_{l+\frac{1}{2}}(x)$ $\Rightarrow$ $R(x)$ 解: $j_l(x) = \sqrt{\frac{\pi}{2x}}J_{l+\frac{1}{2}}(x)$

↑ l 阶球 Bessel 函数

$\rightarrow n_l(x) = \sqrt{\frac{\pi}{2x}}N_{l+\frac{1}{2}}(x) = (-1)^{l+1}\sqrt{\frac{\pi}{2x}}J_{-(l+\frac{1}{2})}(x)$

l 阶球 Neumann 函数

$N_\nu(x) = \frac{\cos\nu\pi J_\nu - J_{-\nu}}{\sin\nu\pi}$, $\nu = l+\frac{1}{2}$ ($\cos\nu\pi = 0$, $\sin\nu\pi \to (-1)^l$)

j_l, n_l 有初等表达式!

$$J_{1/2}(x) = \sum_{k=0}^{\infty}(-1)^k \frac{1}{k!\,\Gamma(k+\frac{3}{2})}\left(\frac{x}{2}\right)^{2k+\frac{1}{2}}$$

$\Gamma(x+1) = x\Gamma(x)$,

$\Gamma(k+\frac{3}{2}) = (k+\frac{1}{2})(k-\frac{1}{2})\cdots\frac{3}{2}\cdot\frac{1}{2}\Gamma(\frac{1}{2})$

$= \frac{1}{2^{k+1}}(2k+1)(2k-1)\cdots 3\cdot 1\sqrt{\pi} = \frac{1}{2^{k+1}}\frac{(2k+1)!}{k!}\sqrt{\pi}$

$(2k+1)$, $(2k-1)$

$2k(2k-2)\cdots 2 = (2k-2)!! = 2^k\cdot k!$

$$= \sum_{k=0}^{\infty}(-1)^k\frac{1}{(2k+1)!}x^{2k+\frac{1}{2}}\sqrt{\frac{2}{\pi}}$$

$$\sin x = \sum_{k=0}^{\infty}(-1)^k\frac{1}{(2k+1)!}x^{2k+1}$$

$\Rightarrow J_{1/2}(x) = \sqrt{\frac{2}{\pi x}}\sin x$

$j_0(x) = \sqrt{\frac{\pi}{2x}}J_{1/2}(x) = \frac{1}{x}\sin x$

同理 $J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}}\cos x = -N_{1/2}(x)$ $\Rightarrow n_0(x) = -\frac{1}{x}\cos x$

0阶 → l阶 递推

$n_l(x)$ $\frac{d}{dx}(x^\nu J_\nu) = x^\nu J_{\nu-1} \Rightarrow \frac{1}{x}\frac{d}{dx}(x^\nu J_\nu) = x^{\nu-1} J_{\nu-1}$

$\Rightarrow \left(\frac{1}{x}\frac{d}{dx}\right)^l (x^\nu J_\nu) = x^{\nu-l} J_{\nu-l}$

$\nu = -\frac{1}{2}$ $\left(\frac{1}{x}\frac{d}{dx}\right)^2 g(x) = \frac{1}{x}\frac{d}{dx}\left[\frac{1}{x}\frac{dg}{dx}\right]$

$\Rightarrow \left(\frac{1}{x}\frac{d}{dx}\right)^l (x^{-\frac{1}{2}} J_{-1/2}) = x^{-\frac{1}{2}-l} J_{-(l+\frac{1}{2})}$

$\Rightarrow J_{-(l+\frac{1}{2})}(x) = x^{l+\frac{1}{2}} \left(\frac{1}{x}\frac{d}{dx}\right)^l \left(\sqrt{\frac{2}{\pi}}\frac{1}{x}\cos x\right) = \sqrt{\frac{2}{\pi}} x^{l+\frac{1}{2}} \left(\frac{1}{x}\frac{d}{dx}\right)^l \left(\frac{\cos x}{x}\right)$

$\Rightarrow n_l(x) = (-1)^{l+1} x^l \left(\frac{1}{x}\frac{d}{dx}\right)^l \left(\frac{\cos x}{x}\right)$

特例: $n_0 = -\frac{\cos x}{x}$, $n_1 = -\frac{\cos x + x\sin x}{x^2}$

$j_l(x)$ $\frac{d}{dx}(x^{-\nu} J_\nu) = -x^{-\nu} J_{\nu+1} \Rightarrow \frac{1}{x}\frac{d}{dx}(x^{-\nu} J_\nu) = -x^{-(\nu+1)} J_{\nu+1}(x)$

$\Rightarrow \left(\frac{1}{x}\frac{d}{dx}\right)^l (x^{-\nu} J_\nu) = (-1)^l x^{-(\nu+l)} J_{\nu+l}$

$\nu = \frac{1}{2} \Rightarrow \left(\frac{1}{x}\frac{d}{dx}\right)^l (x^{-\frac{1}{2}} J_{1/2}) = (-1)^l x^{-(l+\frac{1}{2})} J_{l+\frac{1}{2}}$

$\Rightarrow J_{l+\frac{1}{2}}(x) = (-1)^l \sqrt{\frac{2}{\pi}} x^{l+\frac{1}{2}} \left(\frac{1}{x}\frac{d}{dx}\right)^l \left(\frac{\sin x}{x}\right)$

$\Rightarrow j_l(x) = \sqrt{\frac{\pi}{2x}} J_{l+\frac{1}{2}} = (-1)^l x^l \left(\frac{1}{x}\frac{d}{dx}\right)^l \left(\frac{\sin x}{x}\right)$

$j_0(x) = \frac{\sin x}{x}$, $j_1(x) = \frac{\sin x - x\cos x}{x^2}$

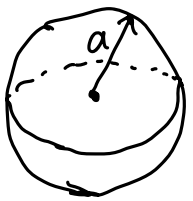
渐近行为 $x\to 0$: $j_0(x)\to 1$, $j_1(x), j_2(x)\cdots \to 0$

$n_0(x), n_1(x), n_2(x)\cdots \to \infty$

$x\to\infty$: $j_0, j_1, j_2\cdots \to 0$

$n_0, n_1, n_2\cdots$

应用:



$\frac{\partial u}{\partial t}=k\nabla^2 u,\quad u|_{r=a}=0,\quad u|_{t=0}=f(r,\theta,\varphi)$

$u=v(r,\theta,\phi)T(t)\Rightarrow \frac{T'}{kT}=\frac{\nabla^2 v}{v}=-\lambda\Rightarrow \nabla^2 v+\lambda v=0$ ← Helmholtz equation 球坐标下分离变量得球Bessel方程

本征值问题: $\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\left[\lambda-\frac{l(l+1)}{r^2}\right]R=0$ (角向)

$R(0)<\infty,\quad R(a)=0$

$\Rightarrow \frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+[\lambda r^2-l(l+1)]R=0$. S-L理论

权函数 $\rho(r)=r^2$

$x=\sqrt{\lambda}r \Longrightarrow \frac{d}{dx}\left(x^2\frac{dR}{dx}\right)+[x^2-l(l+1)]R=0$

$\Rightarrow R=j_l(x)=j_l(\sqrt{\lambda}r)$.

~~$n_l(x)\xrightarrow{x\to 0}\infty$~~

$j_l(x)$ 第n个正零点

$R(a)=0\Rightarrow j_l(\sqrt{\lambda}a)=0\Rightarrow \sqrt{\lambda_n}a=\mu_n^{(l)}$

本征值 $\lambda_n=\left(\frac{\mu_n^{(l)}}{a}\right)^2$,

本征函数 $j_l\left(\frac{\mu_n^{(l)}}{a}r\right)$, $n=1,2,3\cdots$

例: $l=0$: $j_0(x)=\frac{1}{x}\sin x\Rightarrow \mu_n^{(0)}=n\pi$, $n=1,2,3\cdots$

$$u(r,\theta,\phi,t)=\sum_{l=0}^{\infty}\sum_{m=0}^{l}\sum_{n=1}^{\infty} e^{-k\left(\frac{\mu_n^{(l)}}{a}\right)^2 t} j_l\left(\frac{\mu_n^{(l)}}{a}r\right)\cdot\left[A_{lm}\cos m\phi + B_{lm}\sin m\phi\right]\cdot P_l^m(\cos\theta)$$

$u\big|_{t=0}=f(r,\theta,\phi)$，由正交性定系数

↑

$\cos m\phi$, $\sin m\phi$, P_l^m, j_l 均有正交性

广义傅里叶级数

$[0,a]$上：$f(r)=\sum_{n=1}^{\infty} C_n j_l\left(\frac{\mu_n^{(l)}}{a}r\right)$

$$C_n=\frac{1}{\left\|j_l\left(\frac{\mu_n^{(l)}}{a}r\right)\right\|^2}\int_0^a f(r)\, j_l\left(\frac{\mu_n^{(l)}}{a}r\right) r^2\,dr$$

↑ 权函数

↗

$$\int_0^a\left[j_l\left(\frac{\mu_n^{(l)}}{a}r\right)\right]^2 r^2\,dr$$

↖ 初等函数积分，可直接计算

虚宗量球Bessel (不作要求)

$\hat{j}_l(ix), n_l(ix)$ 满足 $(ix)^2\frac{d^2\hat{j}_l(ix)}{d(ix)^2}+2(ix)\frac{d\hat{j}_l(ix)}{d(ix)}+[(ix)^2-l(l+1)]\hat{j}_l(ix)=0$

$$\Rightarrow x^2\frac{d^2\hat{j}_l(ix)}{dx^2}+2x\frac{d\hat{j}_l(ix)}{dx}+[-x^2-l(l+1)]\hat{j}_l(ix)=0$$

定义
$$\hat{i}_l(x)=\sqrt{\frac{\pi}{2x}}I_{l+\frac{1}{2}}(x)\propto\sqrt{\frac{\pi}{2x}}J_{l+\frac{1}{2}}(ix)\propto\hat{j}_l(ix)$$
$$k_l(x)=\sqrt{\frac{\pi}{2x}}K_{l+\frac{1}{2}}(x)$$

满足 modified spherical Bessel equation

$$x^2y''(x)+2xy'(x)-[x^2+l(l+1)]y(x)=0$$

$\hat{j}_l(x), n_l(x)$ 类似三角函数，$\hat{i}_l(x), k_l(x)$ 类似指数函数.

球坐标下解 $(\nabla^2-k^2)u=0$ 会遇到
↖ 和通常 Helmholtz eq. 差负一

一阶PDE的特征线解法（不作要求）

一阶PDE $a(x,t)u_x + b(x,t)u_t = c(x,t)$

$a(x,t)u_x + u_t = c(x,t)$

求合适新变量 $\xi(x,t)$, $\eta(x,t)$ 以减少PDE里最高阶偏导的个数

$$\Rightarrow a(u_\xi \xi_x + u_\eta \eta_x) + b(u_\xi \xi_t + u_\eta \eta_t) = c$$

$$\Rightarrow (a\xi_x + b\xi_t)u_\xi + (a\eta_x + b\eta_t)u_\eta = c(\xi,\eta)$$

令 $a\xi_x + b\xi_t = 0$

一般解法

$$\frac{dx}{dt} = \frac{a}{b} \Rightarrow \phi(x,t) = K \quad \text{(通解)}$$

取 $\xi = \phi(x,t)$ 即可

特征线，(x,t)平面上的一族曲线

积分常数

$(a\eta_x + b\eta_t)u_\eta = c$

令 $\eta = t$ ← 最简做法，η 取法不唯一

$b u_\eta = c$ u_ξ 消失，PDE简化

证明：$\phi(x,t) = K \Rightarrow \phi_x dx + \phi_t dt = 0 \Rightarrow \frac{dx}{dt} = -\frac{\phi_t}{\phi_x}$

但 $\phi(x,t) = K$ 是 $\frac{dx}{dt} = \frac{a}{b}$ 之通解 $\Rightarrow \frac{a}{b} = -\frac{\phi_t}{\phi_x} \Rightarrow a\phi_x + b\phi_t = 0$

取 $\xi = \phi(x,t)$ 即满足 $a\xi_x + b\xi_t = 0$

例1: $a=\text{const}.\ b=1,\ c=0.$ $au_x+u_t=0 \Rightarrow u_\eta=0$

$$\begin{cases}\xi=x-at,\\ \eta=t\end{cases}$$

$\Rightarrow u=f(\xi)=f(x-at)$

$\frac{dx}{dt}=a \Rightarrow x=at+K \Rightarrow x-at=K$

$\xi=\phi(x,t)=x-at$

例2: $xu_x+u_t=A$

$a=x \quad b=1 \quad c=A$

$a\xi_x+b\xi_t=0$

$\frac{dx}{dt}=x \Rightarrow x=Ke^t \Rightarrow xe^{-t}=K \Rightarrow \xi=xe^{-t}$

$\eta=t$ (η取法随意)

$\Rightarrow u_\eta=A \Rightarrow u=A\eta+f(\xi)$ "积分常数"

$=At+f(xe^{-t})$

二阶PDE分类，标准形式

典型二阶PDE		二次曲线
波动方程 $\frac{\partial^2 u}{\partial t^2}-a^2\frac{\partial^2 u}{\partial x^2}=0$	← 双曲型	$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$
热传导 $\frac{\partial u}{\partial t}-k\frac{\partial^2 u}{\partial x^2}=0$	← 抛物型	$y=x^2$
Laplace $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$	← 椭圆型	$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

即使方程右边≠0，只要不含二阶导，不影响分类，例如

$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=2\frac{\partial u}{\partial x}+3u$ 为椭圆型

二次曲线 $ax^2+2bxy+cy^2+dx+ey+f=0$

$$\Rightarrow a\underbrace{\left(x+\frac{b}{a}y\right)}_{=\xi}{}^2+\left(c-\frac{b^2}{a}\right)\underbrace{y^2}_{=\eta}+\cdots=0 \Rightarrow a^2\xi^2-\Delta\eta^2+\cdots=0$$

$\Delta=b^2-ac$

$\Delta\begin{cases}>0 & \text{双曲线}\\ =0 & \text{抛物线}\\ <0 & \text{椭圆}\end{cases}$

系数一般是 (x,y) 的函数

二阶PDE: $aU_{xx}+2bU_{xy}+cU_{yy}+dU_x+eU_y+fU+g=0$

$U_{xx}=\frac{\partial^2 u}{\partial x^2}$，$U_{xy}=\frac{\partial^2 u}{\partial x\partial y}$

分类？化简？

a,b,c,d,e,f,g 为常数

配方

常系数情况 $au_{xx}+2bu_{xy}+cu_{yy}=\Phi(u,u_x,u_y)$

设 $a\neq 0 \Rightarrow a\left(\frac{\partial}{\partial x}+\frac{b}{a}\frac{\partial}{\partial y}\right)^2 u+\left(c-\frac{b^2}{a}\right)\frac{\partial^2 u}{\partial y^2}=\Phi$

$\left(\frac{\partial}{\partial x}+\frac{b}{a}\frac{\partial}{\partial y}\right)\left(\frac{\partial}{\partial x}+\frac{b}{a}\frac{\partial}{\partial y}\right)$ $=u_{xx}+2\frac{b}{a}u_{xy}+\frac{b^2}{a^2}u_{yy}$

(ξ,η), $\frac{\partial}{\partial\xi}=\frac{\partial}{\partial x}+\frac{b}{a}\frac{\partial}{\partial y}=x_\xi\frac{\partial}{\partial x}+y_\xi\frac{\partial}{\partial y}$

$\frac{\partial}{\partial\eta}=\frac{\partial}{\partial y}=x_\eta\frac{\partial}{\partial x}+y_\eta\frac{\partial}{\partial y}$

Jacobian $\Rightarrow \begin{pmatrix}x_\xi & x_\eta\\ y_\xi & y_\eta\end{pmatrix}=\begin{pmatrix}1 & 0\\ \frac{b}{a} & 1\end{pmatrix}$

可取 $\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}1 & 0\\ \frac{b}{a} & 1\end{pmatrix}\begin{pmatrix}\xi\\ \eta\end{pmatrix}$

PDE $\Rightarrow a u_{\xi\xi}+\left(c-\frac{b^2}{a}\right)u_{\eta\eta}=\Phi$

$u_{\xi\xi}-\frac{\Delta}{a^2}u_{\eta\eta}=\Phi$,

$\Delta=b^2-ac$

$\Delta\begin{cases}>0\\ =0\\ <0\end{cases}$ $(\rho,\sigma)=\left(\xi,\sqrt{\left|\frac{a^2}{\Delta}\right|}\,\eta\right)$ $\Longrightarrow$

$u_{\rho\rho}-u_{\sigma\sigma}=\Phi$, 双曲型

$u_{\rho\rho}=\Phi$ 抛物型

$u_{\rho\rho}+u_{\sigma\sigma}=\Phi$ 椭圆型

标准形式

一般情况

系数是 x, y 的函数

$$a u_{xx} + 2b u_{xy} + c u_{yy} + d u_x + e u_y + f u + g = 0$$

$\Downarrow \xi(x,y), \eta(x,y)$

系数为 (x,y) 的函数

$$A u_{\xi\xi} + 2B u_{\xi\eta} + C u_{\eta\eta} + D u_\xi + E u_\eta + F u + G = 0$$

系数是 (ξ,η) 的函数

Jacobian

$$\begin{pmatrix} u_x \\ u_y \end{pmatrix} = \begin{pmatrix} \xi_x & \eta_x \\ \xi_y & \eta_y \end{pmatrix} \begin{pmatrix} u_\xi \\ u_\eta \end{pmatrix} = J \begin{pmatrix} u_\xi \\ u_\eta \end{pmatrix} \quad \text{即} \quad \begin{aligned} u_x &= \xi_x u_\xi + \eta_x u_\eta \\ u_y &= \xi_y u_\xi + \eta_y u_\eta \end{aligned}$$

$$\Rightarrow u_{xx} = \xi_{xx} u_\xi + \xi_x (u_\xi)_x + \eta_{xx} u_\eta + \eta_x (u_\eta)_x$$

$$= \xi_{xx} u_\xi + \xi_x (\xi_x u_{\xi\xi} + \eta_x u_{\xi\eta}) + \eta_{xx} u_\eta + \eta_x (\xi_x u_{\eta\xi} + \eta_x u_{\eta\eta})$$

$$= (\xi_x)^2 u_{\xi\xi} + 2\xi_x \eta_x u_{\xi\eta} + (\eta_x)^2 u_{\eta\eta} + \xi_{xx} u_\xi + \eta_{xx} u_\eta$$

u_{xy}, u_{yy}, u_x, u_y 同理

$$\Rightarrow \begin{aligned} A &= a(\xi_x)^2 + 2b\xi_x\xi_y + c(\xi_y)^2 \\ B &= a\xi_x\eta_x + b(\xi_x\eta_y + \xi_y\eta_x) + c\xi_y\eta_y \\ C &= a(\eta_x)^2 + 2b\eta_x\eta_y + c(\eta_y)^2 \\ D &= a\xi_{xx} + 2b\xi_{xy} + c\xi_{yy} + d\xi_x + e\xi_y \\ E &= a\eta_{xx} + 2b\eta_{xy} + c\eta_{yy} + d\eta_x + e\eta_y \\ F &= f \\ G &= g \end{aligned}$$

"容易证明"(一番硬算):

新变量下的判别式 $\bar{\Delta} = B^2 - AC = (\xi_x \eta_y - \xi_y \eta_x)^2 \Delta$ ← $\Delta = b^2 - ac$

$> 0 \Rightarrow \bar{\Delta}$ 与 Δ 同号

$\Rightarrow$ Δ 的符号是变量代换下的不变量:

$$\mathrm{sgn}(\Delta) = \begin{cases} 1 & \Delta > 0 \quad \text{双曲型} \\ 0 & \Delta = 0 \quad \text{抛物} \\ -1 & \Delta < 0 \quad \text{椭圆} \end{cases}$$

↑ 了解此事实即可,本页剩余部分可跳过

计算量较少的推导:

$$a u_{xx} + 2b u_{xy} + c u_{yy} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right) \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{pmatrix} u + (u \text{的一阶偏导项})$$

$M = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$

$$= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right) M \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{pmatrix} u + (u \text{的一阶偏导项})$$

$$\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right) \begin{pmatrix} a \frac{\partial u}{\partial x} + b \frac{\partial u}{\partial y} \\ b \frac{\partial u}{\partial x} + c \frac{\partial u}{\partial y} \end{pmatrix}$$

$$= \frac{\partial}{\partial x}\left(a \frac{\partial u}{\partial x} + b \frac{\partial u}{\partial y}\right) + \frac{\partial}{\partial y}\left(b \frac{\partial u}{\partial x} + c \frac{\partial u}{\partial y}\right)$$

$$\begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{pmatrix} = \begin{pmatrix} \xi_x & \eta_x \\ \xi_y & \eta_y \end{pmatrix} \begin{pmatrix} \frac{\partial}{\partial \xi} \\ \frac{\partial}{\partial \eta} \end{pmatrix} = J \begin{pmatrix} \frac{\partial}{\partial \xi} \\ \frac{\partial}{\partial \eta} \end{pmatrix}$$

$$\left(\frac{\partial}{\partial \xi}, \frac{\partial}{\partial \eta}\right) J^T M J \begin{pmatrix} \frac{\partial}{\partial \xi} \\ \frac{\partial}{\partial \eta} \end{pmatrix} + (u \text{的一阶偏导项})$$

$$\bar{M} = \begin{pmatrix} A & B \\ B & C \end{pmatrix} = J^T M J$$

$$\Rightarrow \det \bar{M} = (\det J)^2 \det M \Rightarrow \bar{\Delta} = (\det J)^2 \Delta$$

$\det \bar{M} = -\bar{\Delta}$, $\det M = -\Delta$

$$= (\xi_x \eta_y - \xi_y \eta_x)^2 \Delta$$

特征方程 → 分类 + 标准形式

$$A = a(\xi_x)^2 + 2b\xi_x\xi_y + c(\xi_y)^2 = 0$$ ← 二阶PDE

$$B = a\xi_x\eta_x + b(\xi_x\eta_y + \xi_y\eta_x) + c\xi_y\eta_y$$

$$C = a(\eta_x)^2 + 2b\eta_x\eta_y + c(\eta_y)^2$$

希望二阶偏导项系数A, B, C中有些为0, 以简化方程

定理 若 $\phi(x,y)=K$ (K: 常数) 是一阶ODE $a\left(\frac{dy}{dx}\right)^2 - 2b\frac{dy}{dx} + c = 0$ (特征方程)

的一个通解(即由K标记的一族积分曲线)

则 $\xi = \phi(x,y)$ 是一阶PDE $a(\xi_x)^2 + 2b\xi_x\xi_y + c(\xi_y)^2 = 0$

的一个特解.

例: $a=9,\ b=0,\ c=-4,\ 9u_{xx} - 4u_{yy} = 0$

ODE: $9\left(\frac{dy}{dx}\right)^2 - 4 = 0 \Rightarrow 3\frac{dy}{dx} \pm 2 = 0 \Rightarrow 2x \pm 3y = K$

$2x+3y=K_1 \xrightarrow{\text{根据定理}} \xi = 2x+3y$ 满足 $9(\xi_x)^2 - 4(\xi_y)^2 = 0$

$2x-3y=K_2$ 同理.

证明: $\phi(x,y)=K \Rightarrow \phi_x dx + \phi_y dy = 0 \Rightarrow \frac{dy}{dx} = -\frac{\phi_x}{\phi_y}$

$\phi(x,y)=K$ 是特征方程之解

$$\Rightarrow a\left(-\frac{\phi_x}{\phi_y}\right)^2 - 2b\left(-\frac{\phi_x}{\phi_y}\right) + c = 0$$

$$\Rightarrow a(\phi_x)^2 + 2b\phi_x\phi_y + c(\phi_y)^2 = 0$$

取 $\xi = \phi(x,y) \Rightarrow a(\xi_x)^2 + 2b\xi_x\xi_y + c(\xi_y)^2 = 0$

特征方程 $\Rightarrow \frac{dy}{dx} = \frac{b \pm \sqrt{b^2-ac}}{a} = \frac{b \pm \sqrt{\Delta}}{a}$

(1) $\Delta = b^2 - ac > 0$ ← 特征线

$\frac{dy}{dx} = \frac{b+\sqrt{\Delta}}{a} \Rightarrow \phi(x,y) = K_1 \Rightarrow \xi = \phi(x,y) \Rightarrow A = 0$

$\frac{dy}{dx} = \frac{b-\sqrt{\Delta}}{a} \Rightarrow \psi(x,y) = K_2 \Rightarrow \eta = \psi(x,y) \Rightarrow C = 0$

$\Rightarrow A = C = 0$, 由于 $B^2 - AC = \Delta \neq 0 \Rightarrow B \neq 0$

$\Rightarrow u_{\xi\eta} = \Phi_1(\xi, \eta, u, u_\xi, u_\eta)$ ← 双曲型 PDE 的标准形式

$\Downarrow \rho = \xi + \eta, \ \sigma = \xi - \eta$

$\Rightarrow u_{\rho\rho} - u_{\sigma\sigma} = \Phi_2(\rho, \sigma; u, u_\rho, u_\sigma)$

(2) $\Delta < 0 \Rightarrow \frac{dy}{dx} = \frac{b \pm i\sqrt{|\Delta|}}{a}$

$\frac{dy}{dx} = \frac{b+i\sqrt{|\Delta|}}{a} \Rightarrow \phi(x,y) = K_1 \Rightarrow \xi = \phi(x,y) \Rightarrow A = 0$

$\frac{dy}{dx} = \frac{b-i\sqrt{|\Delta|}}{a} \Rightarrow \psi(x,y) = K_2 \Rightarrow \eta = \psi(x,y) \Rightarrow C = 0$

→ 两方程互为复共轭 $\Rightarrow$ 可取 $\psi(x,y) = \phi^*(x,y)$

$\begin{cases} \xi = \phi(x,y) \\ \eta = \phi^*(x,y) \end{cases} \Rightarrow u_{\xi\eta} = \Phi(\xi, \eta, u, u_\xi, u_\eta)$

$\rho=\xi+\eta$, $\sigma=i(\xi-\eta)$ (实数) $\Rightarrow \frac{\partial}{\partial\xi}=\frac{\partial}{\partial\rho}+i\frac{\partial}{\partial\sigma}$, $\frac{\partial}{\partial\eta}=\frac{\partial}{\partial\rho}-i\frac{\partial}{\partial\sigma}$

$\Rightarrow \left(\frac{\partial}{\partial\rho}+i\frac{\partial}{\partial\sigma}\right)\left(\frac{\partial}{\partial\rho}-i\frac{\partial}{\partial\sigma}\right)u=\Phi$

$\Rightarrow u_{\rho\rho}+u_{\sigma\sigma}=\Phi$

此方程和 $u_{\xi\eta}=\Phi$ 都称为椭圆型 PDE 标准形式

(3) $\Delta=0 \Rightarrow \frac{dy}{dx}=\frac{b}{a} \Rightarrow \phi(x,y)=k \Rightarrow \xi=\phi(x,y) \Rightarrow A=0$

$\downarrow$ sgn(Δ) 在变量代换下不变

$\overline{\Delta}=B^2-AC=0$, $A=0$ $\Rightarrow B=0 \Rightarrow C\neq 0$ (否则不是二阶 PDE)

$\Rightarrow u_{\eta\eta}=\Phi(\xi,\eta,u,u_\xi,u_\eta)$

η 可任选 (和 ξ 独立) $\det\begin{pmatrix}\xi_x & \xi_y\\ \eta_x & \eta_y\end{pmatrix}\neq 0$

得到 PDE 标准形式后，可进一步求解方程. 例如，可利用 Riemann 方法求解. 参见 A. Sommerfeld《物理学中的偏微分方程》或 J. Ockendon et. al《应用偏微分方程》. (均有中文译文)

例: $x^2u_{xx}-2xyu_{xy}+y^2u_{yy}+xu_x+yu_y=0$

$\Rightarrow \Delta=(-xy)^2-x^2y^2=0 \Rightarrow$ 抛物型

$a(\frac{dy}{dx})^2-2b\frac{dy}{dx}+c=0 \Rightarrow \frac{dy}{dx}=\frac{b}{a}=\frac{-xy}{x^2}=-\frac{y}{x}$

$\Rightarrow \frac{dy}{y}+\frac{dx}{x}=0 \Rightarrow \ln y+\ln x=k' \Rightarrow xy=k$

取 $\xi=xy$, η 可任取, 取 $\eta=y$

$\Rightarrow C=y^2=\eta^2$

$E=y=\eta \quad D=F=G=0$

$\Rightarrow \eta^2u_{\eta\eta}+\eta u_\eta=0 \Rightarrow \eta u_{\eta\eta}+u_\eta=0$

$v=u_\eta \Rightarrow \eta v_\eta+v=0 \xrightarrow{\eta=e^t} v=\varphi(\xi)e^{-t}=\frac{\varphi(\xi)}{\eta}$

η 的积分常数换成 ξ 的任意函数 $\varphi(\xi)$

$\Rightarrow u_\eta=v=\frac{\varphi(\xi)}{\eta} \Rightarrow u=\varphi(\xi)\ln|\eta|+\psi(\xi)$

$=\varphi(xy)\ln|y|+\psi(xy)$

Tricomi eq. $yu_{xx}+u_{yy}=0$, $\Delta=-y$

(1) $y>0 \Rightarrow$ 椭圆型

$y(\frac{dy}{dx})^2+1=0 \Rightarrow \frac{dy}{dx}=\pm\frac{i}{\sqrt{y}} \Rightarrow x\pm i\frac{2}{3}y^{\frac{3}{2}}=K$

取 $\xi=x$, $\eta=\frac{2}{3}y^{\frac{3}{2}}$

用A,B,C,D...公式或直接计算:

$\Rightarrow u_{\xi\xi}+u_{\eta\eta}+\frac{1}{3\eta}u_\eta=0$

(2) $y=0$. (3) $y<0$. 略.

常系数二阶PDE可消去一阶偏导项

$$au_{xx}+2bu_{xy}+cu_{yy}+du_x+eu_y+fu+g=0$$

$$u=e^{\lambda x+\mu y}v(x,y)$$

λ,μ待定

$$\Rightarrow a(v_{xx}+2v_x\lambda+\lambda^2 v)e^{\lambda x+\mu y}+2b(v_{xy}+v_x\mu+v_y\lambda+\mu\lambda)e^{\lambda x+\mu y}$$
$$+c(v_{yy}+2v_y\mu+\mu^2 v)e^{\lambda x+\mu y}$$
$$+d(v_x+\lambda v)e^{\lambda x+\mu y}+e(v_y+\mu v)e^{\lambda x+\mu y}+fve^{\lambda x+\mu y}+g=0$$

$$\Rightarrow av_{xx}+2bv_{xy}+cv_{yy}+(2\lambda a+2\mu b+d)v_x+(2\lambda b+2\mu c+e)v_y$$
$$+(a\lambda^2+2\lambda\mu b+c\mu^2+\lambda d+\mu e+f)v+ge^{-\lambda x-\mu y}=0$$

一阶偏导项$=0$ $\Leftarrow$ $\begin{cases} v_x\text{系数} & 2\lambda a+2\mu b+d=0 \\ v_y\text{系数} & 2\lambda b+2\mu c+e=0 \end{cases}$

$$\Downarrow$$

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix}\begin{pmatrix} 2\lambda \\ 2\mu \end{pmatrix}=\begin{pmatrix} -d \\ -e \end{pmatrix}$$

$$\Downarrow$$

$$\begin{pmatrix} \lambda \\ \mu \end{pmatrix}=\frac{1}{2}\begin{pmatrix} a & b \\ b & c \end{pmatrix}^{-1}\begin{pmatrix} -d \\ -e \end{pmatrix}=\frac{1}{2\Delta}\begin{pmatrix} c & -b \\ -b & a \end{pmatrix}\begin{pmatrix} d \\ e \end{pmatrix}$$

即 $\lambda=\dfrac{cd-be}{2\Delta}$，$\mu=\dfrac{ae-bd}{2\Delta}$